U-I-001159A (DSA-DM)
March 13, 1975
SUPERSEDING
Revision of
Int. Fed. Spec.
U-I-001159 (DSA-DM)
March 9, 1966

INTERIM FEDERAL SPECIFICATION

IMPRESSION MATERIAL, DENTAL, RUBBER-BASE PASTE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers polysulfide rubber-base dental impression material, suitable for taking oral impressions.

1.2 Classification. Polysulfide material shall be of the following classes, as specified (see 6.1):

Class 1. Heavy body.
Class 2. Regular body.
Class 3. Light body.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal
U-S-746 Stone, Artificial, Dental
PPP-B-566 Boxes, Folding, Paperboard.

FSC 6520

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PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.
PPP-B-676 Boxes, Setup.

STANDARDS

Federal

FED-STD-123 Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, DC.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATION

Military

MIL-L-10547 Liners, Case and Sheet, Overwrap; Water-Vaporproof of Waterproof, Flexible

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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"2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION (ADA)

ADA Specification No. 19 Elastomeric Impression Material

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611.)

3. REQUIREMENTS

3.1 Material. Material shall be of a polysulfide rubber base which, when mixed with an accelerator in accordance with instructions (see 3.5) accompanying the package, produces a free-flowing elastic material suitable for taking impressions in the mouth.

3.2 Description. Dental rubber base impression material shall be furnished in a package containing a tube of rubber base paste, a tube of accelerator paste, two keys, one for each tube, a mixing pad, and a set of instructions for use. The polysulfide rubber base and the accelerator shall be of contrasting colors and supplied in airtight collapsible metal composition tubes fabricated of material that will not react nor contaminate the impression materials. In addition, the tubes shall be of sufficient strength and flexibility that leakage will not occur during normal use. Shall be furnished in class 1, class 2, or class 3, as specified (see 6.1). Class 1 shall be heavy-bodied and highly viscous for use in the impression tray to drive home the light-bodied impression material (class 3). Class 2 shall be regular bodied and intermediate in viscosity. Class 3 shall be light bodied and free flowing suitable for use in a syringe.

3.2.1 Accessories. Keys, one for each tube, shall be furnished with each unit package to facilitate the extrusion of impression materials. In addition, each unit package shall contain a mixing pad compatible for use with mixing polysulfide base impression materials. The pads shall contain a minimum of 35 sheets, cut in sheets 4 by 5 inches (in.) $\pm$ 1/8 in. and 0.008 in. $\pm$ 0.002 inch in thickness. The sheets shall be glued on two adjacent sides. As an alternate, pad sheets shall measure 4-1/2 by 6-1/4 in. $\pm$ 1/8 in. and shall be 0.008 in. $\pm$ 0.002 in. in thickness. Each pad of 35 sheets shall be mounted on a suitable cardboard base.

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3.3 Physical properties. Physical properties shall be in accordance with the requirements of table I.

Table I - Physical Properties

Requirements	Class I	Class 2	Class 3
Mixing time (minutes maximum)	1	1	1
Working time (minutes)	2-1/4 to 4-3/4	1-3/4 to 4-1/4	4-1/4 to 6-3/4
Consistency - disk diameter (millimeters)	30 ± 3	37 ± 3	44 ± 3
Permanent deformation (maximum percent)	4.0	4.0	4.0
Dimensional stability - Maximum change at 24 hours (percent)	0.40	0.40	0.40
Strain in compression (percent)	4.0 t to 20.0	4.0 to 20.0	4.0 to 20.0
Rate of gain of hardness-			
a. Minutes to reach a Durometer of 40, maximum	10		
b. Minutes to reach a Durometer of 25, maximum		10	
c. Minutes to reach a Durometer of 18, maximum			18
Flow (percent), maximum	2.0	2.0	2.0

3.3.1 Odor and flavor. The impression material shall have no objectionable odor or flavor.

3.3.2 Toxicity. The supplier shall furnish prior to delivery, a statement directly to the Directorate of Medical Materiel, Technical Services Branch, (Code ATTS-2), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, that the impression material, used according to the directions supplied with the material, has no toxic effect on a normal, healthy person.

3.3.3 Reproduction. The material shall reproduce a groove 0.025 millimeter (mm.) in width, when tested in accordance with 4.4.11.

3.3.4 Compatibility.

3.3.4.1 Compatibility with gypsum. The material shall impart a smooth surface to and separate cleanly from hardened dental stone, complying with U-S-746 (Setting time: 10 ± 3 minutes). A stone cast poured against the material in accordance with 4.4.12.1 shall reproduce the line 0.025 mm. wide. The dental stone shall itself be capable of reproducing the line 0.025 mm. wide when allowed to harden against the test block (see figure 1).

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3.3.4.2 Compatibility with metalizing baths. The materials shall be compatible with alkaline silver-plating baths when tested as specified in 4.4.12.2.

3.3.5 Deterioration. The permanent deformation shall not exceed 4.0 percent for material which has been stored at 60 ± 2.0 degrees Centigrade (°C.) for 7 days when tested as specified in 4.4.13.

3.4 Amount of material in unit containers. Class 2 regular and class 3 light-bodied impression material shall be furnished in one tube of base and one tube of catalyst in each unit package. Class 1 heavy-bodied impression material shall be furnished in two tubes of base and one tube of catalyst in each unit package. The minimum volume of mix shall be 120 milliliters (ml.) for regular (class 2) and light-bodied (class 3) material. The minimum volume of mix shall be 165 ml. for heavy-bodied (class 1) material.

3.5 Instructions for use. Adequate and accurate instructions for manipulating the material shall accompany each unit package. These instructions shall include the following minimum requirements:

1. The proportioning of base and accelerating materials to give a proper mix by weight or extruded length.

2. The mixing time required to produce a streak-free mix.

3. Procedure for working the paste and leaving the impression in the mouth.

4. Precautionary statement on finger (or skin) and clothing contact, and removal of stains.

3.6 Shelf life. The shelf life of the impression material shall be no less than 24 months when stored at controlled room temperature (59 to 86 degrees Fahrenheit (°F.)).

3.7 Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the impression material to the date of delivery to the Government.

3.8 Marking. Each tube and unit package of material shall be marked with a serial number or combination of letters and numbers which shall refer to the manufacturer's records for that particular lot or batch. The year and month of manufacture shall be suitably coded as part of the lot number. In addition, each tube of polysulfide rubber base shall be permanently and legibly identified as "Heavy", "Regular" or "Light" according to the impression material class. Each tube of accelerator shall be permanently and legibly identified as "Catalyst" or identified as accelerator. The unit package shall be permanently and legibly identified "Heavy-Bodied", "Regular-Bodied", or "Light-Bodied" according to the applicable class of material.

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3.9 Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the brand of impression material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association (ADA) Specification No. 19 for elastomeric impression material, type I, polysulfide base. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.10 Workmanship. Workmanship shall be first class throughout. The base and the accelerator shall be homogeneous, uniform in consistency and free of defects, foreign materials, and impurities which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, IV and V (see 4.2.1, 4.3 and 4.3.1) and table III (see 4.2.2).

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be one package of impression material.

Table II Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor	S-1	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with table III. Unit of product for test purposes shall be one package of inspection material.

Table III Tests

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations per lot or batch no.
Mixing time	3.3	4.4.3	1	1 1/	1
Working time	3.3	4.4.4	3	3 2/	9
Consistency	3.3	4.4.5	1	3 2/	3
Permanent deformation	3.3	4.4.6	1	3 2/	3
Strain in compression	3.3	4.4.7	1	3 2/	3
Dimensional stability	3.3	4.4.8	1	3 2/	3
Flow	3.3	4.4.9	1	3 2/	3
Rate of gain of hardness	3.3	4.4.10	1	3 2/	3
Reproduction	3.3.3	4.4.11	1	2 1/	2
Compatibility with gypsum	3.3.4.1	4.4.12.1	1	2 1/	2
Compatibility with metalizing baths	3.3.4.2	4.4.12.2	1	2 1/	2
Deterioration	3.3.5	4.4.13	1	3 2/	3
Material volume	3.4	S.M.I.	3	1 1/	3

1/ A lot shall be rejected if any determination fails the test requirement.
2/ A lot shall be rejected if any of the averages of the determinations per unit of product fails the test.

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4.2.3 Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. Impression material shall be examined for defects, including, but not limited to, those listed in table IV. In process examination may be used for examination of these defects.

Table IV. Classification of Defects

Categories	Defects
Major	
101	Package not complete with base, accelerator, keys for tubes, mixing pad, and instructions.
102	Tubes not properly identified.
103	Tubes not airtight and collapsible.
104	Odor or flavor objectionable.
105	Base and accelerator not of contrasting colors.
Minor:	
201	Instructions not complete or legible.
202	Mixing pads not furnished with minimum number of sheets specified.
203	Dimensions of sheets on mixing pad not as specified.

4.3.1 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table V. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.4 Tests.

4.4.1 Standard testing conditions. All physical tests shall be made under uniform atmospheric conditions 23.0 ± 2.0° C. and 50 ± 10 percent relative humidity, unless otherwise specified. Equipment and material shall be conditioned in the testing room for not less than 10 hours prior to making the tests.

4.4.2 Standard mixes. Two sizes of standard mixes shall be used depending on the size of the test specimen to be made. A small mix shall consist of amounts of base and accelerator to give 8 ± 0.5 gram (gm.) of the mixed material. The large mix shall consist of amounts of base and accelerator to give 20 ± 1 gm. of the mixed material.

4.4.3 Mixing time. Impression material shall be spatulated in accordance with furnished instructions and the time noted to produce a homogenous, smooth mix, free from lumps, granules and color streaks.

4.4.4 Working time. A small standard mix shall be prepared and retained in a compact layer approximately 3 mm. thick. A 10 gm. steel rod, 2.4 mm. in diameter (see figure 2), shall be applied to the surface at 20 second intervals and raised slowly. The needle shall be withdrawn from the test material at the same rate (no faster than) at which the needle penetrated the material. The end of the rod shall be wiped clean after each application. The working time shall be recorded as the time elapsed from the start of the mix to the time at which the material fails to string and an indentation persists when the rod is withdrawn. Three determinations shall be made and the average working time shall be reported to the nearest 30 seconds.

4.4.5 Consistency. A volume of 0.5 ml. of a small standard mix of material shall be deposited on a cellophane covered glass plate. Two minutes after the start of mix, the 0.5. ml. of material shall be covered with another cellophane sheet and a plate glass (approximately 75 gm.) plus a 500-gm. weight. At 12 minutes after the start of mix the load (glass and weight) shall be removed and the major and minor diameters of slumped mass of material shall be measured. The average value for three specimens, recorded to the nearest millimeter, shall be recorded as the consistency.

4.4.6 Permanent deformation.

4.4.6.1 Preparation of specimens. Test specimens shall be made from a small standard mix by placing a metal ring 30 mm. inside diameter and 16 mm. high on a flat glass or metal plate and by filling the ring slightly more than one-half full with impression material mixed according to the manufacturer's

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instructions. A metal or polymethyl methacrylate mold 12.7 mm. inside diameter, 25.4 mm. outside diameter and 19 mm. high shall be placed immediately inside the ring and shall be forced into the material until the mold touches the plate and the material has exuded onto the top of the mold. A flat glass, metal or polymethyl methacrylate plate shall be pressed onto the top of the mold to remove excess material. Two minutes after the start of mix, the mold and its accompanying plates shall be placed in a water bath maintained at 37.0 ± 0.1° C. The mold and the plates shall remain in the water bath the minimum time recommended by the manufacturer for leaving the impression material in the mouth (see 3.5). One minute after removal of the specimen from the water bath, the specimen shall be tested by the method described in 4.4.6.2. (The molds may be lubricated with a silicone grease (high-vacuum type) to prevent the impression material adhering to the mold.)

4.4.6.2 Test procedure. The specimen shall be placed in a suitable instrument consisting essentially of a dial gauge, sensitive to 0.02 mm., mounted to a steady base and equipped with a screw, positioned in such a manner that sufficient force can be applied to the specimen to produce the required amount of strain (see figure 3). One minute after removal from the water bath, a lightweight plate, weighing 2.5 ± 0.5 gm. shall be placed on top of the specimen and the foot of the dial gauge allowed to contact the plate. The weight of the plate and the force of the spring in the gauge shall be 50 ± 5 gm. The dial gauge shall be read 30 seconds after the foot of the gauge contacts the plate and the value recorded as reading A. The dial gauge shall be lowered 2.3 mm. by means of the screw and the pressure shall be maintained on the specimen for 30 seconds, then released and the specimen allowed to rest under no load (except that of the lightweight plate) for 30 seconds. The dial gauge foot shall again be put into contact with the plate on the specimen for 30 seconds and a second reading shall be taken. This value shall be recorded as reading B. The difference between readings A and B, divided by the original length of the specimen and multiplied by 100 shall be recorded as the percent permanent deformation. The height of the mold (19.0 mm) shall be considered to be the original length of the specimen. The average permanent deformation of three specimens shall be reported to the nearest 0.1 percent.

4.4.7 Strain in compression.

4.4.7.1 Preparation of specimens. Specimens shall be prepared as described in 4.4.6.1

4.4.7.2 Test procedure. Six minutes after removal from the water bath, the specimen shall be transferred to an appropriate instrument (see figure 4), and shall be subjected to a load calculated to produce a stress of 100 grams per square centimeter (gm. per sq. cm.). Thirty seconds later the dial

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indicator shall be read. This value shall be designated as reading A. Sixty seconds after application of a stress of 100 gm. per sq. cm., an additional load calculated to produce a total stress on the specimen of 1000 gm. per sq. cm., shall be gradually applied during an interval of 10 seconds. Thirty seconds after initiation of the stress of 1000 gm. per sq. cm., a reading of the dial indicator shall be taken. This reading shall be designated as reading B. The difference between readings A and B divided by the original length of the specimen, times 100, shall be considered as the percentage of strain between the stresses of 100 and 1000 gm. per. sq. cm. The original length of the specimen shall be considered as the height of the mold (19.0 mm.) used in forming it. The test shall be performed on three specimens and the average value shall be reported to the nearest 0.1 percent as the percent strain in compression.

4.4.8 Dimensional stability.

4.4.8.1 Preparation of specimens. The mold (see figure 5) shall be lubricated with silicone grease and placed on a flat surface on which two reference crosslines have been ruled. The reference lines shall be approximately 60 degree grooves, 0.05 mm. wide. The two reference lines shall be not less than 4.5 centimeters (cm.) apart. The mold shall be slightly overfilled with impression material from a large standard mix. A sheet of polyethylene shall be placed over the impression material and the excess pressed out with a flat plate. Two minutes from the beginning of mix, the mold and plates shall be placed in a 37.0 $\pm$ 1.0° C. water bath. The mold and plates shall remain in the water bath the minimum time recommended by the manufacturer for leaving the impression material in the mouth (see 3.5). The polyethylene shall be stripped off and the flash trimmed away before separating the halves of the mold.

4.4.8.2 Testing procedure. The specimens shall be dusted with talcum powder on the surface opposite that on which the reference marks appear, and placed on a smooth glass surface which has also been dusted with talcum powder. The distance between the crosslines shall be measured to the nearest 0.01 mm., two minutes after removal from the water bath (reading A). Additional measurements shall be made at 24 hours (reading B) after the initial reading. The difference between the reference lines on the plate shall be recorded as reading C. The difference between readings A and B divided by reading C and multiplied by 100 shall be recorded as the value for percentage of dimensional change. The values for percent dimensional change shall be reported as the average of tests on three specimens recorded to the nearest 0.05 percent.

4.4.9 Flow.

4.4.9.1 Preparation of specimens. Specimens shall be prepared as described in 4.4.6.1, except that upon removal from the mold, the specimens shall be stored at room temperature until one hour from the start of mix.

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4.4.9.2 Test procedure. The specimens shall be placed in an instrument consisting essentially of a dial gauge graduated in 0.02 mm. (see figure 3), mounted to a steady base. At 1 hour from the start of mix, a light-weight plate weighing 2.5 ± 0.5 gm. shall be placed on top of the specimen and the foot of the dial gauge shall be brought into contact with the plate. The force exerted by the dial gauge plus the weight of the plate shall be 100 gm. The dial gauge shall be read 30 seconds after the foot of the gauge contacts the plate and the value recorded as reading A. A second reading shall be taken 15.0 minutes after the foot of the dial gauge contacts the plate and the value recorded as reading B. The difference between reading A and reading B, divided by the original length of the specimen and multiplied by 100 shall be considered to be the percent flow. The height of the mold (19.0 mm.) shall be considered to be the original length of the specimen. The average value for three specimens, recorded to the nearest 0.1 percent shall be recorded as the percent flow.

4.4.10 Rate of gain of hardness.

4.4.10.1 Preparation of specimens. A large standard mix of impression material shall be prepared and retained in an overfilled Durometer cup (see figure 6). The overfill shall be from 1/16 to 1/8 in. thick. The top of the specimen shall be formed flat and parallel to the plate.

4.4.10.2 Test procedure. The hardness characteristics of the impression material shall be tested using a Shore Type A Durometer, or equivalent testing equipment. Hardness readings shall be made at one minute intervals starting as soon as the rubber is sufficiently tack-free to permit testing. The hardness of the impression material is determined by pressing the durometer instrument firmly against the set rubber. Care must be taken to keep the indentor point at least 1/4-inch from the side walls of the cup. The pressure of the instrument against the sample must be applied quickly, and the maximum needle reading noted. The semi-cured rubbers permit considerable needle drift, if a time interval exists, between application of pressure and reading the instrument. The application of pressure and reading the instrument must occur simultaneously. Three readings shall be averaged for each unit of product tested. Defective readings shall be discarded and additional readings taken. A new spot shall be penetrated for each reading. The foot of the instrument shall be wiped clean before each penetration. The total elapsed time from the start of the mix to reach the required hardness shall be reported to the nearest minute.

4.4.11 Reproduction. A small standard mix shall be made in accordance with the manufacturer's instructions. A ring 16 mm. high and approximately 30 mm. inside diameter, shall be positioned on a stainless steel test block similar to that shown in figure 1, so that the intersection of a crossline and line A, 0.025 mm. wide, is in the center of the ring. The stainless

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steel test blocks may be lightly dusted with talcum powder and the excess talcum powder blown off, if the impression material adheres to the blocks. The ring shall be slightly overfilled with impression material. A flat plate shall be placed on top and the excess material shall be squeezed out. Two minutes after the start of mix, the entire assembly shall be placed in a water bath maintained at 37 ± 1° C. The assembly shall remain in the water bath the minimum time recommended by the manufacturer for leaving the impression material in the mouth (3.5). The ring with the impression material shall be separated from the test block immediately and the impression shall be examined under low-angle illumination without magnification. The reproduction shall be considered to be satisfactory if the 0.025 mm. wide groove is continuous for the full inside diameter of the ring.

4.4.12 Compatibility.

4.4.12.1 Compatibility with gypsum. A stone slurry under gentle vibration, shall be poured against the impression made in 4.4.11 within 10 minutes from the time it is separated from the test block. The calcined stone shall be artificial dental stone as specified in 3.3.4.1. The poured impression shall be placed in an air bath at 23.0 + 2.0° C. and 50 ± 10 percent relative humidity for 30 minutes. The stone cast shall be removed and examined without magnification, under low angle illumination. The impression material shall be considered to be compatible with gypsum if the groove 0.025 mm. in width is reproduced continuously for the full inside diameter of the ring.

4.4.12.2 Compatibility with metalizing baths. A large standard mix of impression material shall be made and spread out to form a layer approximately 3 mm. thick. Thirty minutes after mixing, two strips, approximately 25 mm. wide by 50 mm. long, shall be cut from the layer of material. One strip shall be suspended in the prepared copper sulfate solution and the other in the silver cyanide solution. Approximately one half the strip shall be in the solution and one half in air above the solution. After 16 hours immersion in the solutions, the strips shall be removed, rinsed in water, blown dry, with a gentle air blast and inspected. Discoloration and slight softening are acceptable but blistering, wrinkling, gross softening, etching, sluffing or swelling reject the material. After the inspection, a mix of calcined gypsum (artificial dental stone as specified in 3.3.4.1) shall be poured over one side of the test specimen. One hour later the impression material shall be stripped from the set stone. A slight color transfer to the stone is acceptable but a gross transfer of rubber, adhering to the gypsum surface, rejects the material. Test solutions are prepared as follows:

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a. Copper plating.

Copper sulfate	125 gm.
Sulphuric acid	37.5 gm.
Phenol 90%	5 ml.
Ethyl alcohol	25 ml.
Distilled water	500 ml.

b. Silver plating.

Silver cyanide	28 gm.
Potassium cyanide	53 gm.
Potassium carbonate	42 gm.
Distilled water	950 ml.

4.4.13 Deterioration (shelf life). Tubes of base and accelerator shall be stored at 60.0 ± 2.0° C. for 7 days. Permanent deformation tests in accordance with 4.4.6 shall then be run on these materials.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing a tube of rubber base paste, a tube of accelerator paste, two keys, a mixing pad and instructions for use of the class specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit of issue of the class specified shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

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5.2.1 Level A. Thirty six unit packages of one class of material only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B. Thirty six unit packages of one class only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Contract or purchase order number, name of contractor, lot (control) number, date of manufacture (in lieu of date packed) and the following legend shall be shown: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)". In addition, tubes shall be marked as specified in 3.8.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) numbers shall be shown. The date of manufacture shall be shown in lieu of the date packed. Marking shall include the following legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of material required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Acceptable source. Class 1 material shall be "Permlastic Heavy Body" as manufactured by Kerr Manufacturing Co., Division of Sybron Corp. Class 3 material shall be "Permlastic Light Body" as manufactured by Kerr Manufacturing Co., Division of Sybron Corp.

6.3 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with asterisks to indicate where changes from the previous issue have been made.

6.4 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-764-2261	IMPRESSION MATERIAL, DENTAL, Paste, Heavy Bodied: Rubber Base.
6520-00-764-2262	IMPRESSION MATERIAL DENTAL, Paste, Light Bodied: Rubber Base.
6520-00-781-0775	IMPRESSION MATERIAL, Dental, Paste, Regular Bodied: Rubber Base.

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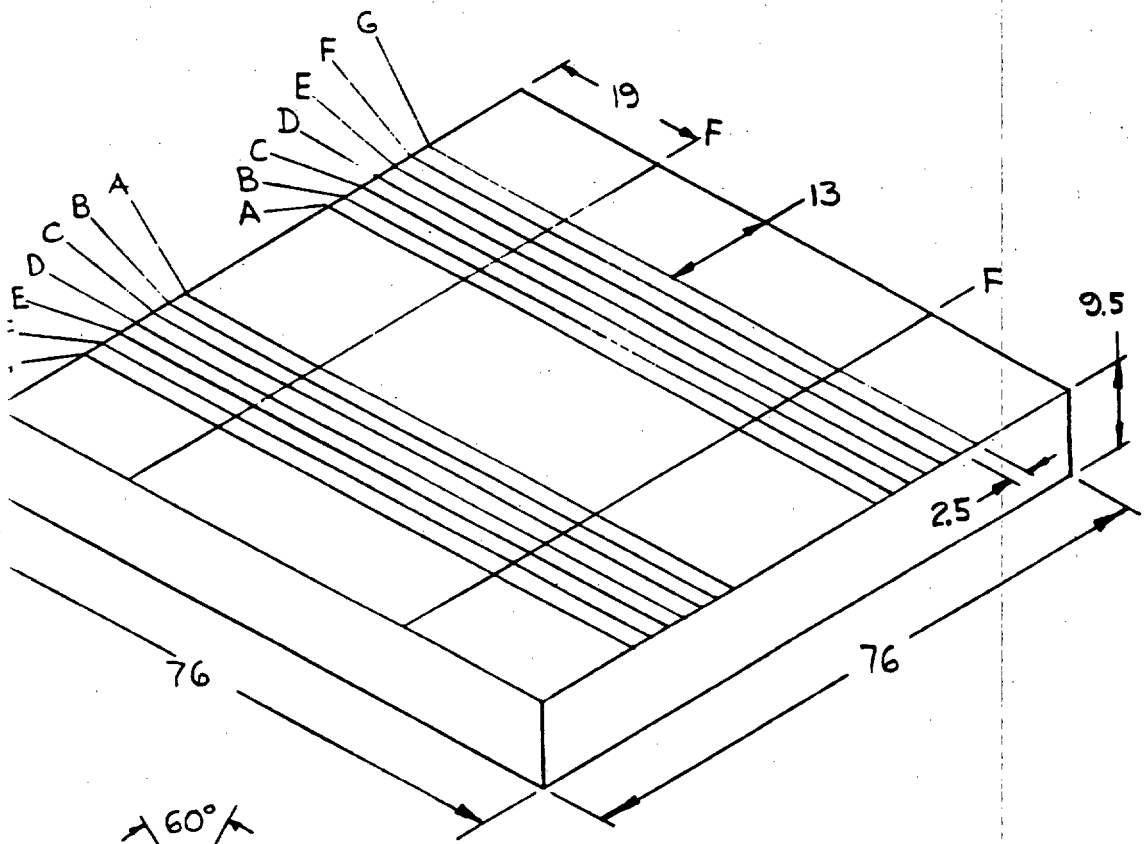
6.5 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with asterisks to indicate where changes from the previous issue have been made.

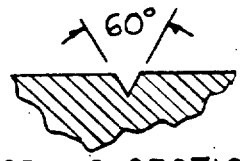
Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1388
is

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CROSS SECTION
OF RULED LINES

ALL DIMENSIONS ARE
IN MILLIMETERS

MILLIMETER
A = 0.025
B = 0.050
C = 0.075
D = 0.100
E = 0.150
F = 0.200
G = 0.300

Figure 1. Test Block for Detail Reproduction

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Adjust length to give 10 gm. weight

NOTE: A suitable non-restraining guide may be used with the needle.

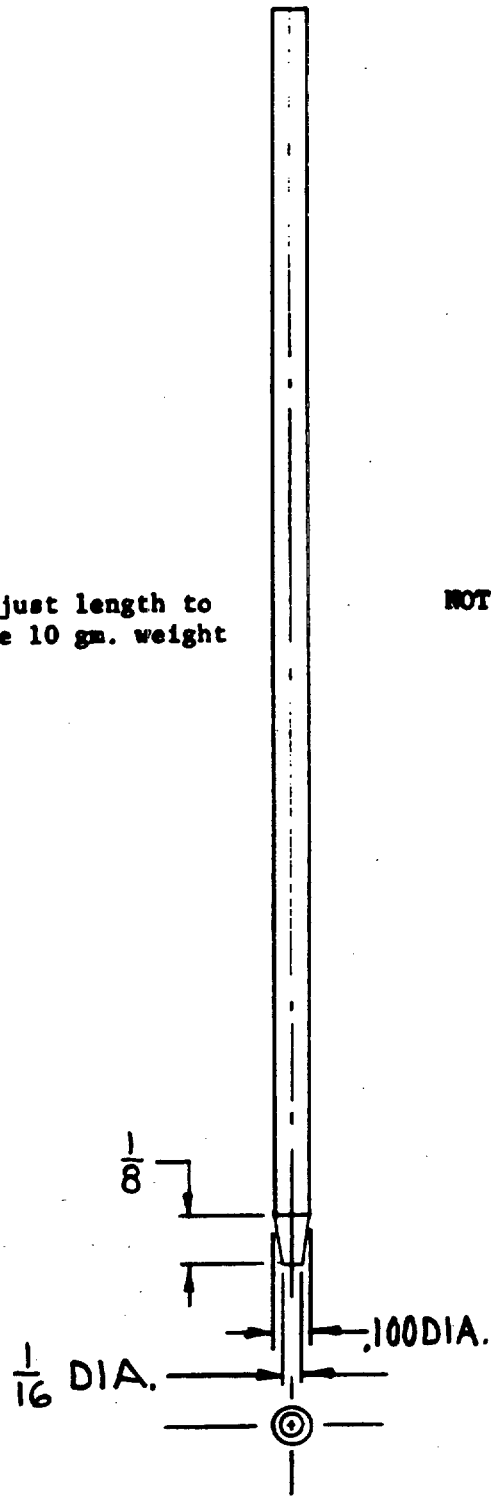


Figure 2. Working Time Test Needle

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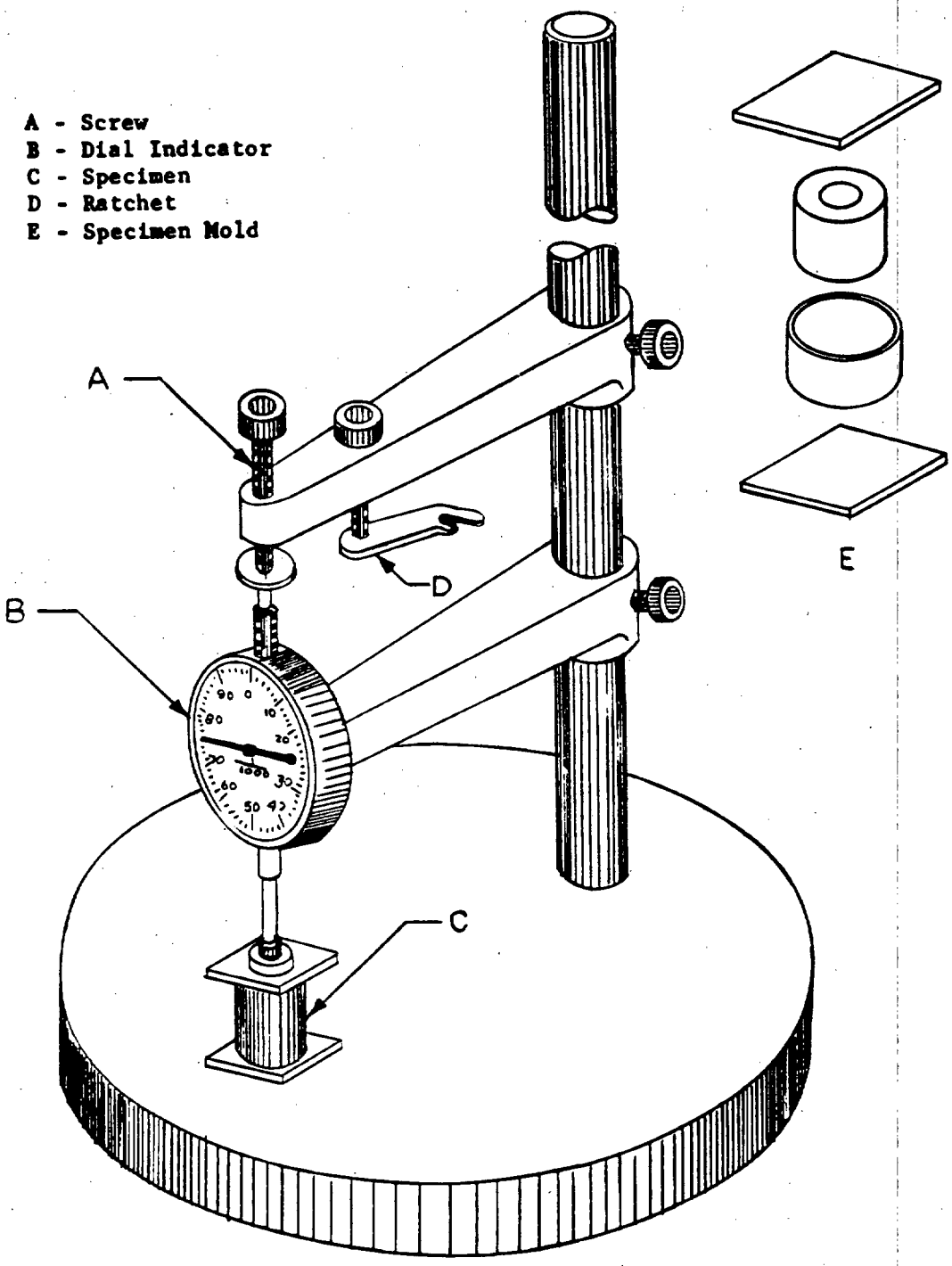


Figure 3. Instrument for Measuring Permanent Deformation

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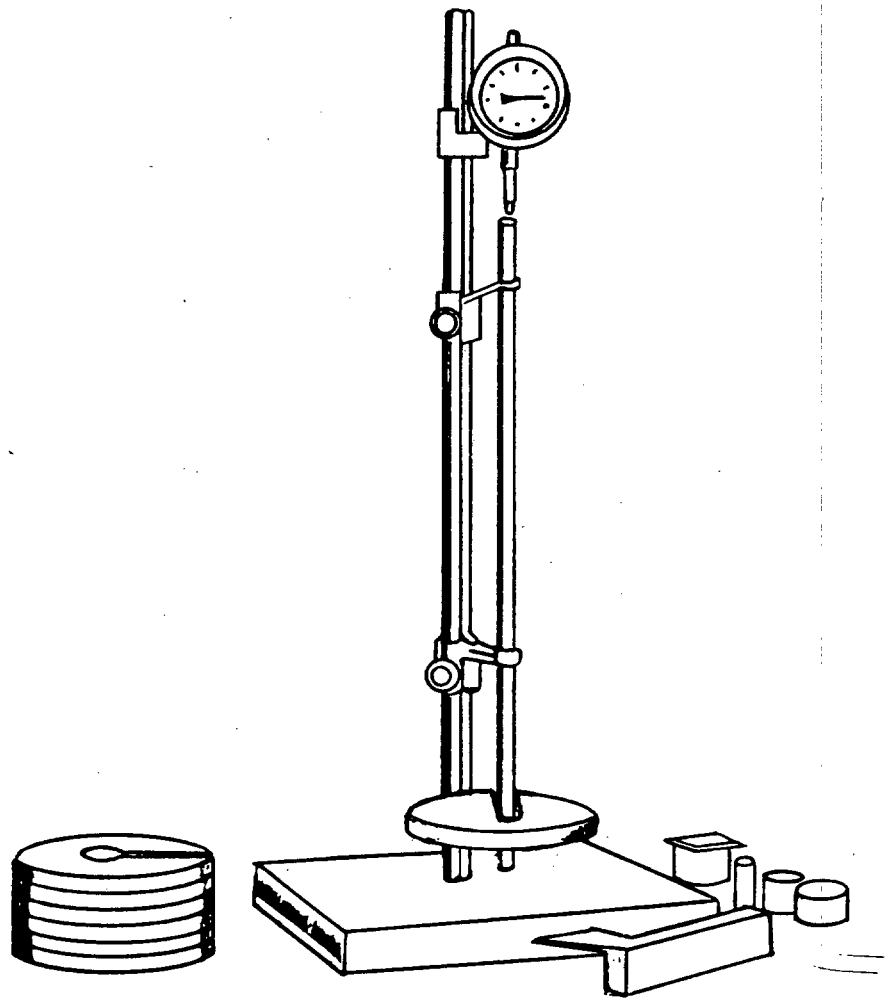
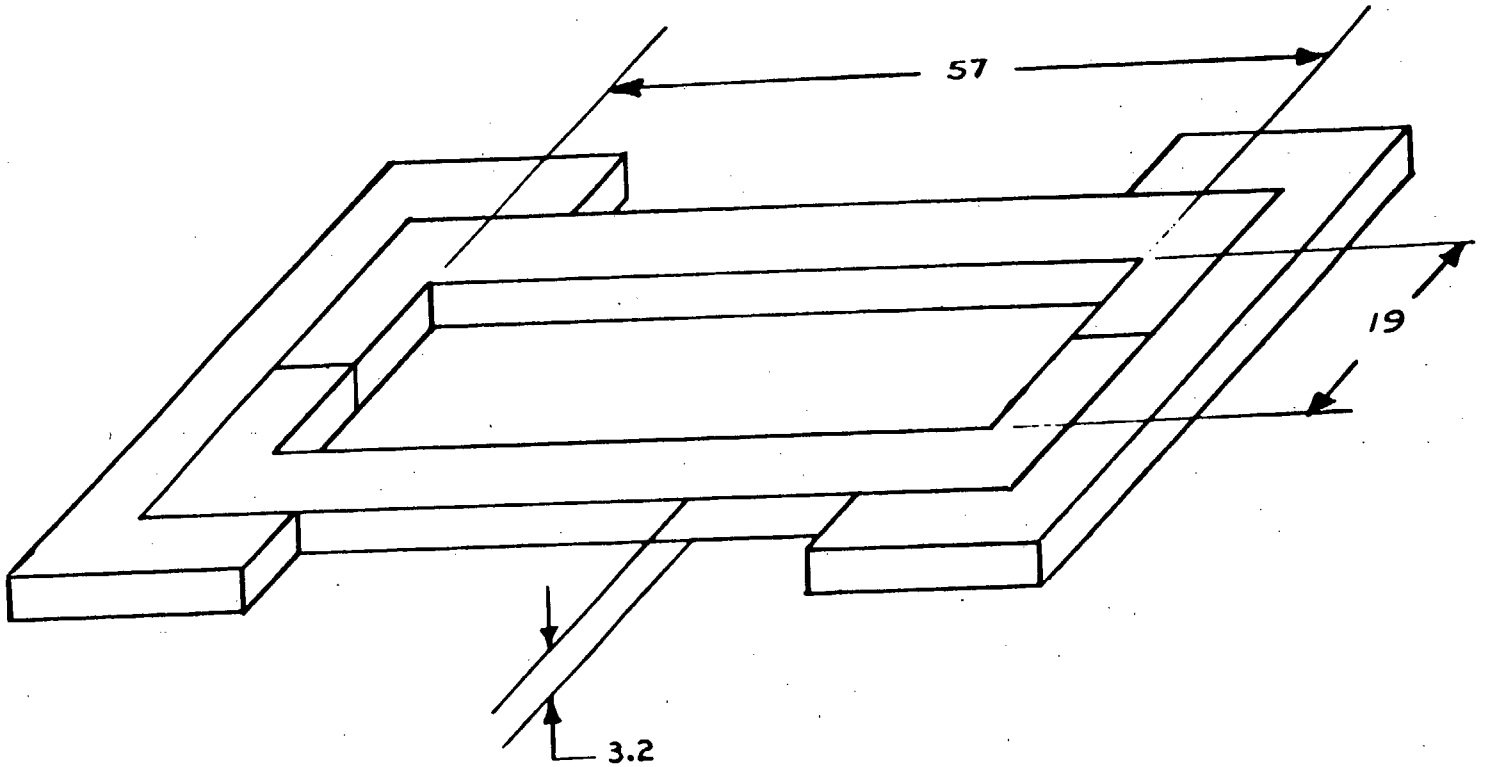


Figure 4. Fixture for Determining Properies in Compression

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All Dimensions In Millimeters

Figure 5. Mold for Dimensional Stability Test

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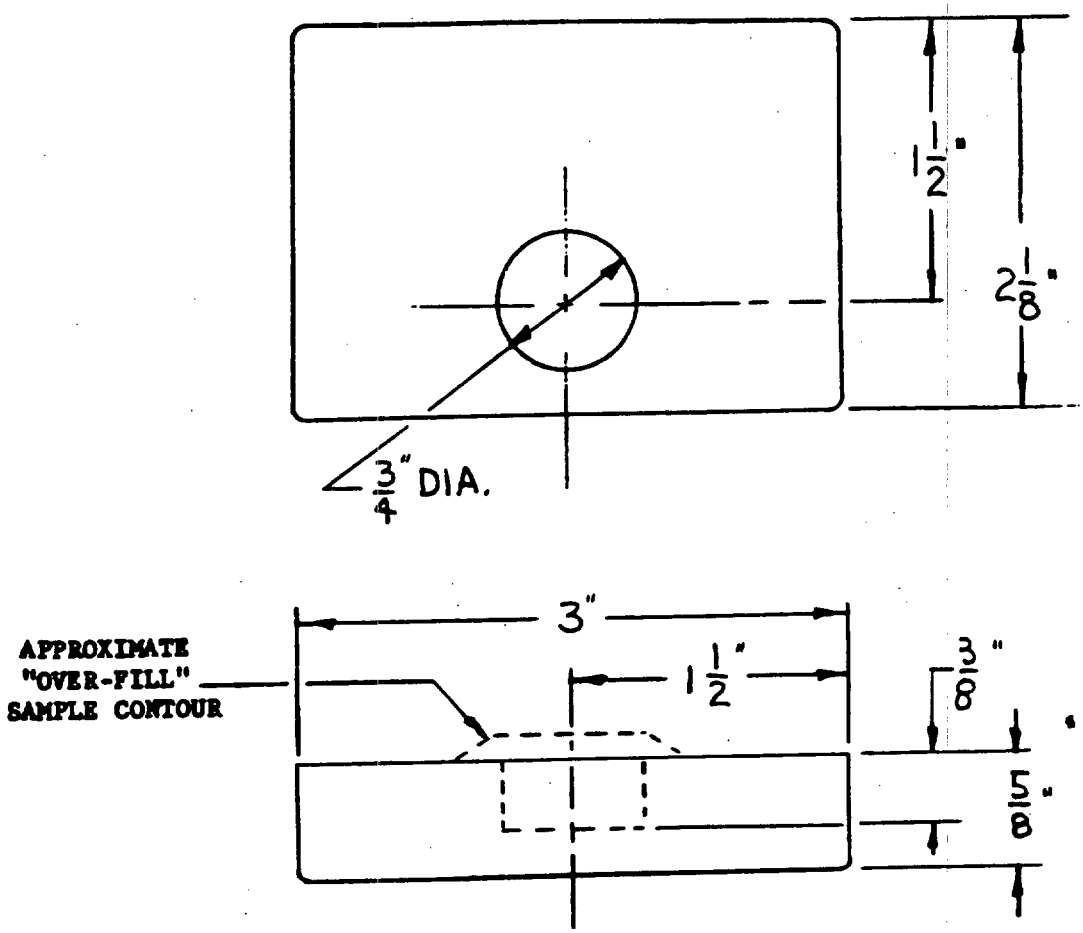


Figure 6. Durometer Test Cup

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
U-I-001159A (DSA-DM) IMPRESSION MATERIAL, DENTAL, RUBBER-BASE PASTE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

☆ GPO: 1974— 603-680/6125 2-1

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